

Experiment# 11

Creating different Types of Inheritance in Classes

Objective

Objective of this lab session is to understand inheritance, different types of inheritance and how to implement inheritance in C++.

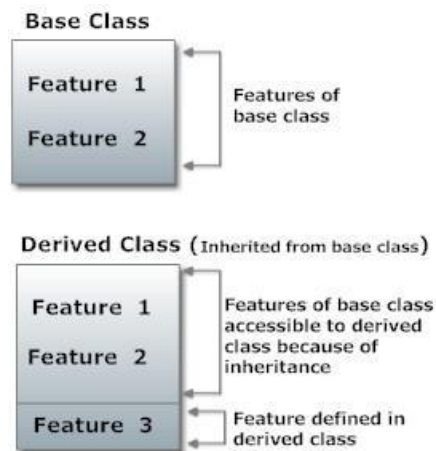
Software Used

Code Blocks/ Dev C++

Theory

Inheritance

Inheritance is one of the key features of object-oriented programming including C++ which allows user to create a new class (**derived class**) from an existing class (**base class**). The derived class inherits all features from a base class and it can have additional features of its own.



Inheritance Visibility Mode

Depending on Access modifier used while inheritance, the availability of class members of Super class in the sub class changes. It can either be private, protected or public.

There are three types of inheritance in C++

1) Public Inheritance

This is the most used inheritance mode. In this the protected member of super class becomes protected members of sub class and public becomes public.

```
class Subclass : public Superclass
```

2) Private Inheritance

In private mode, the protected and public members of super class become private members of derived class.

```
class Subclass : Superclass // By default its private inheritance
```

3) Protected Inheritance

In protected mode, the public and protected members of Super class become protected members of Sub class.

```
class subclass : protected Superclass
```

Table showing all the Visibility Modes

Base class	Derived Class		
	Public Mode	Private Mode	Protected Mode
Private	Not Inherited	Not Inherited	Not Inherited
Protected	Protected	Private	Protected
Public	Public	Private	Protected

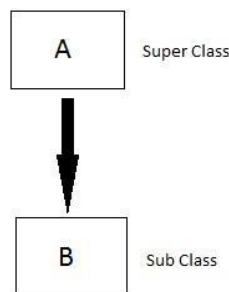
Types of Inheritance

In C++, we have different types of Inheritance. Namely,

- Single Inheritance
- Multiple Inheritance
- Hierarchical Inheritance
- Multilevel Inheritance
- Hybrid Inheritance (also known as Virtual Inheritance)

Single Inheritance

In this type of inheritance one derived class inherits from only one base class. It is the simplest form of Inheritance.



Example#1:

```
#include <iostream>
```

```
using namespace std;
```

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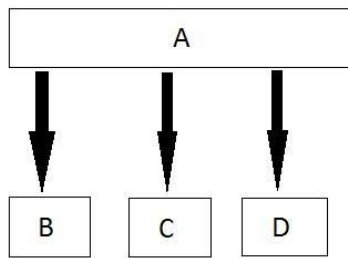
```
// Base class
class Parent
{
int a;
public:
    void setA(int w)
    {
        a = w;
    }
};

// Derived class
class Child: public Parent
{
public:
    void display()
    {
        cout<<a;
    }
};

int main(void)
{
    Child c;
    c.setA(4);
    c.display();
    return 0;
}
```

Hierarchical Inheritance

In this type of inheritance, multiple derived classes inherit from a single base class.

**Example #3**

```

#include <iostream>
using namespace std;
// Base class
class Parent
{
int a;
public:
    void setA(int w)
    {
        a = w;
    }
};
class Child1: public Parent
{
public:
    void getA()
    {
        cout<<a+10<<endl;
    }
};

// Derived class
class Child2: public Parent

```

```

{
    public:
        void getA2()
        {
            cout<<a*10<<endl;
        }
};

int main(void)
{
    Child1 c1;
    c1.setA(4);
    c1.getA();
    Child2 c2;
    c2.setA(5);
    c2.getA2();
    return 0;
}

```

Lab Tasks

1. Create and Implement program in C++ to define a parent class Cell Phone which has the following features Model Number, Screen size, and Camera.(choose data type according to the field).
Implement the following functions to Cell Phone class
 - displayInfo(): to show the cell phone details
 - setInfo(): to set the cell phone details
 - makeCall (int number): to make a call to the number entered by the user.
 - sendSMS(int number, string message): to send sms to number entered by user.

Define the child/derived class of Nokia which has custom attributes other than defined in parent class. For example a simple cell phone cannot send MMS. Make an attribute in child class names has MMSSupport and write a function to send MMS if it is capable of sending MMS (hasMMSSupport=true). Modern phones have internet facility and they have good memory (i.e. RAM, ROM) as well.

2. Create and Implement program in C++ to define a Class Shape which has fields of x and y in it.

Define following functions

set_x()

set_y()

Inherit Class Rectangle (with data members i.e. length and width) Ellipse (with data members i.e. minor_axis and major_axis) and Triangle (with data members i.e. base and height).

Define custom fields of child classes i.e. area ()

Display area, length and width of all child classes.

(Note: area of rectangle= length * width

Area of Ellipse= π * major_axis *minor_axis

Area of triangle= $\frac{1}{2}$ * base * height)

Conclusion

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